AP BIOLOGY TOPIC 7.8

Continuing Evolution: Resistance, Pathogens & Genomic Change

UNIT 7: NATURAL SELECTION (13–20% EXAM WEIGHT)

EVO-3.A | **ANTIBIOTIC RESISTANCE** | **EMERGING PATHOGENS**

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 7.8 (CONTINUING EVOLUTION: RESISTANCE, PATHOGENS & GENOMIC CHANGE)

CED Core Vocabulary

Populations Continue to Evolve — All extant species continue to evolve across generations driven by mutation, natural selection, genetic drift, and gene flow as environments constantly shift.

Gene Duplication — Unequal crossing over creates extra copies of a gene; one copy maintains original function while the duplicate is free to mutate and evolve a novel function...

Fossil Record Shows Ongoing Change — Exhibits both gradualism (slow, continuous morphological shifts) and punctuated equilibrium (long stasis interrupted by rapid bursts of evolution).

Herbicide Resistance in Weeds — Glyphosate (Roundup) selects for weeds with EPSPS enzyme gene amplification or target-site mutations.

Point & Chromosomal Mutations — Spontaneous replication errors continuously introduce new alleles.

Horizontal Gene Transfer (HGT) in Prokaryotes — Bacteria rapidly acquire resistance or virulence genes via conjugation (R-plasmids), transformation, and transduction .

Pesticide Resistance in Insects — Repeated spraying of DDT or pyrethroids selects for insects with pre-existing mutations in voltage-gated sodium channels or detoxifying cytochrome P450 enzymes.

Chemotherapy Resistance in Tumor Cells — Genetically heterogeneous cancer cells within a tumor undergo clonal selection; cells with pre-existing resistance mutations survive chemotherapy and cause relapse.

§2. 1. EVOLUTION IS AN ONGOING PROCESS IN ALL POPULATIONS

Continuous Change

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

1. Evolution Is an Ongoing Process in All Populations ➔

Populations Continue to Evolve ➔ **Fossil Record Shows Ongoing Change**

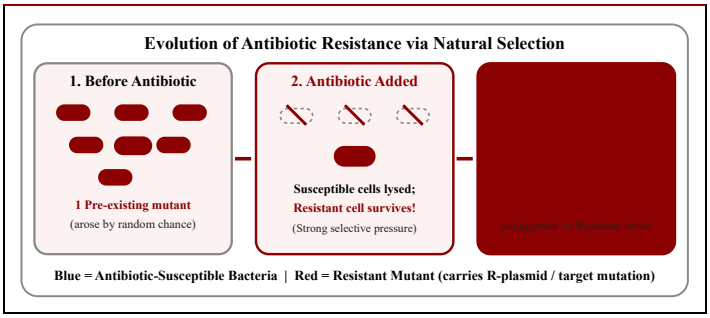
- **Populations Continue to Evolve:** All extant species continue to evolve across generations driven by *mutation, natural selection, genetic drift, and gene flow* as environments constantly shift.
- **Genomic Changes Continuously Generate New Variation:**
  - **Point & Chromosomal Mutations:** Spontaneous replication errors continuously introduce new alleles.
  - **Gene Duplication:** Unequal crossing over creates extra copies of a gene; one copy maintains original function while the duplicate is free to mutate and evolve a **novel function** (e.g., alpha/beta/fetal globin gene family; olfactory receptor genes).
  - **Horizontal Gene Transfer (HGT) in Prokaryotes:** Bacteria rapidly acquire resistance or virulence genes via **conjugation (R-plasmids), transformation, and transduction**.
- **Fossil Record Shows Ongoing Change:** Exhibits both **gradualism** (slow, continuous morphological shifts) and **punctuated equilibrium** (long stasis interrupted by rapid bursts of evolution).

[EXAMPLE & BIOLOGICAL APPLICATION] **Real-Time Field Evolution —**

Darwin's Finches (Peter & Rosemary Grant): On Daphne Major (Galápagos), a severe 1977 drought wiped out small soft seeds, leaving only large, hard *Tribulus* seeds. Medium ground finches (*Geospiza fortis*) with deeper, stronger beaks survived and reproduced, shifting average beak depth upward in just **one generation**!

§4. VISUALIZING POSITIVE SELECTION FOR ANTIBIOTIC RESISTANCE

Pre-Existing Mutation



§5. 3. RAPID PATHOGEN EVOLUTION & EMERGING DISEASES

Viruses & Zoonoses

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

3. Rapid Pathogen Evolution & Emerging Diseases ➔

Why Pathogens Evolve So Rapidly ➔ **Zoonotic Spillover (Emergent Diseases)**

- **Why Pathogens Evolve So Rapidly:**
  - **Extremely Short Generation Times:** Hours/minutes in bacteria and viruses → thousands of generations per year.
  - **High Mutation Rates (Especially RNA Viruses):** Viral RNA polymerases and reverse transcriptases **lack 3′→5′ proofreading exonuclease activity**, generating frequent point mutations.
- **Antigenic Drift vs. Antigenic Shift (Influenza / SARS-CoV-2 / HIV):**
  - **Antigenic Drift:** Gradual accumulation of point mutations in surface glycoprotein genes (e.g., hemagglutinin *H* & neuraminidase *N*, or spike protein) that allow the virus to evade host memory antibodies (why seasonal flu vaccines must be updated annually!).
  - **Antigenic Shift:** Sudden reassortment of RNA genome segments when two different viral strains co-infect the same host cell (e.g., avian + human flu in a pig), creating a novel pandemic strain.
- **Zoonotic Spillover (Emergent Diseases):** New human diseases arise when animal pathogens mutate to bind human cell-surface receptors, cross the species barrier, and adapt to human-to-human transmission.

[EXAMPLE & BIOLOGICAL APPLICATION] **Major Zoonotic Spillover**

Examples:

- **HIV-1:** Crossed from chimpanzees (SIV) → humans; evolves drug resistance rapidly inside a single patient unless treated with multi-drug **HAART** therapy.
- **SARS-CoV-2 & Ebola:** Originated in bats → spilled over into human populations.
- **H1N1 / H5N1 Influenza:** Reassortment between avian (birds) and swine (pigs) reservoirs.

§3. 2. EVOLUTION OF RESISTANCE TO HUMAN CHEMICAL AGENTS

Directional Selection

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

2. Evolution of Resistance to Human Chemical Agents ➔

directional selective pressure ➔ **Pesticide Resistance in Insects** ➔

Herbicide Resistance in Weeds ➔ **Chemotherapy Resistance in Tumor Cells**

- Human application of chemical agents acts as an intense **directional selective pressure** on target populations:
- **Antibiotic Resistance in Bacteria (e.g., MRSA, MDR-TB):**
  - Before antibiotic exposure, a few bacterial cells already possess random mutations (or R-plasmids) conferring resistance (e.g., *β-lactamase enzyme* or efflux pumps).
  - Antibiotic treatment kills susceptible bacteria, eliminating competition and allowing resistant bacteria to survive, reproduce by binary fission, and dominate the population.
- **Pesticide Resistance in Insects:** Repeated spraying of DDT or pyrethroids selects for insects with pre-existing mutations in voltage-gated sodium channels or detoxifying cytochrome P450 enzymes.
- **Herbicide Resistance in Weeds:** Glyphosate (Roundup) selects for weeds with EPSPS enzyme gene amplification or target-site mutations.
- **Chemotherapy Resistance in Tumor Cells:** Genetically heterogeneous cancer cells within a tumor undergo clonal selection; cells with pre-existing resistance mutations survive chemotherapy and cause relapse.

[AP EXAM FRQ SCORING STRATEGY] **CRITICAL AP FRQ WARNING — Avoid Teleological Language!**

NEVER write that bacteria "developed," "mutated in order to," or "learned to resist" antibiotics to survive! Always state: **(1) Random mutations conferring resistance ALREADY existed in the population prior to treatment; (2) The antibiotic acted as a selective pressure killing susceptible cells; (3) Resistant cells had higher fitness, survived, and passed resistance genes to offspring.**

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§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 7.8 [MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]

Topic 7.8: Continuing Evolution: Resistance, Pathogens & Genomic Change ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔

Experimental Perturbation (±2 SE_x̄) ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 1, 2 & 3 (Artificial Selection, Hardy-Weinberg Mathematical Modeling & BLAST Bioinformatics): Comparing allele frequency shifts across generations under directional selection vs. small-population genetic drift simulations, and constructing maximum-parsimony cladograms from BLAST percentage sequence identity matrices (Cytochrome c / β-globin). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Golden Rule: Hardy-Weinberg Equilibrium (HWE): Allele frequencies $p + q = 1$; Genotype frequencies p^2 (homozygous dominant) + 2pq (heterozygous carriers) + q^2 (homozygous recessive) = 1. Golden Calculation Sequence: ALWAYS begin with the recessive phenotype frequency (q^2), take $\sqrt{(q^2)}$ to find q, compute $p = 1 - q$, and then calculate 2pq (heterozygote carrier frequency). Never take the square root of the dominant phenotype frequency ($p^2 + 2pq$)! **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Evolutionary & Biomedical Disruption Case Study: Incomplete antibiotic regimens or agricultural antibiotic overuse exert strong directional selection on bacterial populations (e.g., MRSA or MDR-Tuberculosis): susceptible cells die while rare, pre-existing resistant mutants (arising via random mutation or R-plasmid conjugation) survive and reproduce. In conservation biology, severe population bottlenecks (e.g., Northern Elephant Seals, African Cheetahs) drastically erode genetic diversity via genetic drift, increasing homozygosity for deleterious recessive alleles and vulnerability to novel pathogens.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 7.8 to prove why the claim holds.